How big a chance? I don’t know, and nor does anyone else. My intuition tells me it’s unlikely, but my intuition is informed by a decade and a half of working within the conventional framework of Planck-sized strings and Planck-sized extra dimensions. Perhaps my instincts are old-fashioned. Thankfully, the question will be settled without the slightest concern for anyone’s intuition. If the strings are big, or if some of the extra dimensions are big, the implications for upcoming experiments are spectacular.

In the next chapter, we’ll consider a variety of experiments that will test, among other things, the possibilities of comparatively large strings and large extra dimensions, so here I will just whet your appetite. If strings are as large as a billionth of a billionth (10^{-18}) of a meter, the particles corresponding to the higher harmonic vibrations in Figure 12.4 will not have enormous masses, in excess of the Planck mass, as in the standard scenario. Instead, their masses will be only a thousand to a few thousand times that of a proton, and that’s low enough to be within reach of the Large Hadron Collider now being built at CERN. If these string vibrations were to be excited through high-energy collisions, the accelerator’s detectors would light up like the Times Square crystal ball on New Year’s Eve. A whole host of never-before-seen particles would be produced, and their masses would be related to one another’s much as the various harmonics are related on a cello. String theory’s signature would be etched across the data with a flourish that would have impressed John Hancock. Researchers wouldn’t be able to miss it, even without their glasses.

Moreover, in the braneworld scenario, high-energy collisions might even produce—get this—miniature black holes. Although we normally think of black holes as gargantuan structures out in deep space, it’s been known since the early days of general relativity that if you crammed enough matter together in the palm of your hand, you’d create a tiny black hole. This doesn’t happen because no one’s grip—and no mechanical device—is even remotely strong enough to exert a sufficient compression force. Instead, the only accepted mechanism for black hole production involves the gravitational pull of an enormously massive star’s overcoming the outward pressure normally exerted by the star’s nuclear fusion processes, causing the star to collapse in on itself. But if